

A Sensing Module for Subsurface-level Wildfire Monitoring

Summary

To address the growing wildfire risk to Canada’s Wildland–Urban Interface (WUI), we focused on detecting subsurface “zombie” (also known as: smouldering, overwintering) fires to improve the resilience of community and privately owned structures. An environmental monitoring IoT framework was selected to guide the design and prototyping process. The RFP was rescoped to prioritize data collection in high-risk areas, based on research identifying peat soils as a major source of zombie fires. Community research, including insights from Indigenous communities and the BC Wildfire Service, informed stakeholder needs and design priorities. The team diverged and converged on concepts capable of collecting relevant subsurface peat data using structured ideation and decision tools, resulting in a feasible concept aligned with stakeholder needs and technical requirements.

1 Design Process

This document walks through the design process illustrated in Figure 1. Throughout the entire design process was Team Fundamentals, documents which managed our team’s progress. It includes the [Responsibility Tracker](#), [Team Charter](#), [Engineering Notebook](#) and [Work Log](#) (click to view). All numbers in the image caption are also clickable and direct the reader to the corresponding artifact or artifact folder.

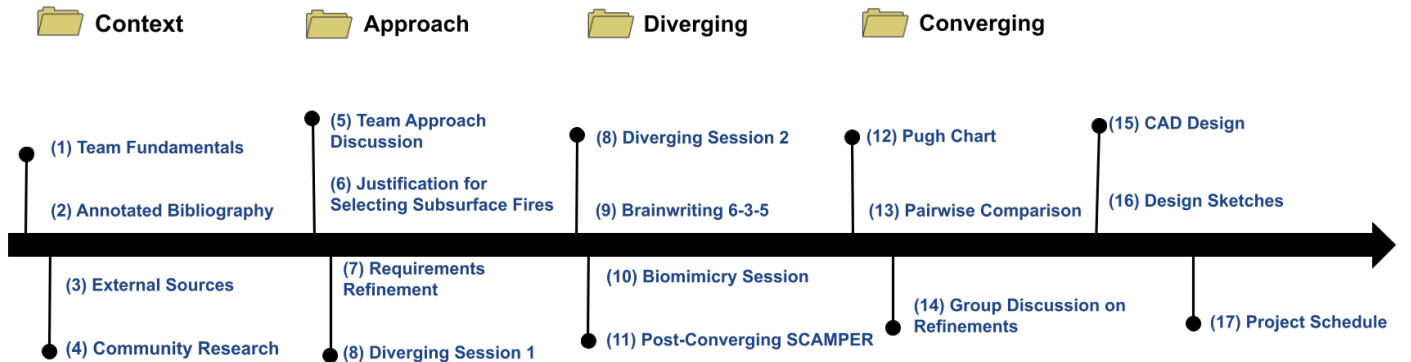


Figure 1: Timeline artifacts displaying key activities throughout the design process. The artifacts can be accessed through the following direct links: [01](#), [02](#), [03](#), [04](#), [05](#), [06](#), [07](#), [08](#), [09](#), [10](#), [11](#), [12](#), [13](#), [14](#), [15](#), [16](#), [17](#).

2 Context and Approach (Click: [Annotated Bibliography](#), [External Sources](#))

The first week was focused on researching wildfires, which highlighted that they are increasingly becoming year-round events. This shift is largely due to overwintering “zombie fires,” which can persist as subsurface smouldering combustion throughout winter in peat-rich boreal soils and re-emerge in spring, increasing early-season ignition risk for wildland–urban interface (WUI) communities [2, 1]. Because they burn underground, they are difficult to detect and monitor using conventional methods. Our research indicates that zombie fires primarily occur in peat soils [2]. They may result from the smouldering remains of surface

wildfires that heat peat soil and become insulated by snow, or from R-tipping, in which rising subsurface temperatures exceed a critical threshold that enables spontaneous ignition [2]. Burn depth typically remains below 50, cm, although it can vary. These fires can also exhibit characteristic gas signatures, such as CO and CO₂ at ratios closer to 1:1 (rather than ~10:1 for flaming combustion), providing measurable indicators despite the difficulty of direct detection [2].

2.1 Stakeholders (Click: [Community Research](#))

Our stakeholders include (1) communities at the WUI, (2) government and policy makers, and (3) forests and the natural environment. A meeting with the BC Wildfire Service confirmed that wildfire activity now persists year-round due to overwintering “zombie fires,” which are difficult and time-consuming to detect because they burn underground without visible flame [5]. These fires intensify subsequent fire seasons through a positive feedback loop. Research also highlighted disproportionate impacts on Indigenous communities such as Fort McKay First Nation, Fort McMurray First Nation, and Fort Nelson First Nation, where repeated evacuations contribute to significant mental health burdens and long-term social disruption [5].

Updated NGOs (Click: [Updated Requirements](#))

The information from the Context and Approach, as well as the Community Research, directly influenced the following updated NGOs. For example, O-1.1 calls for the detection of CO and CO₂ concentrations [Ann. Bib. 6]. Similarly, O-4.1 discusses the requirement to withstand temperatures up to 500°C in the presence of fire, which may occur under smouldering or flaming conditions [Ann. Bib. 5].

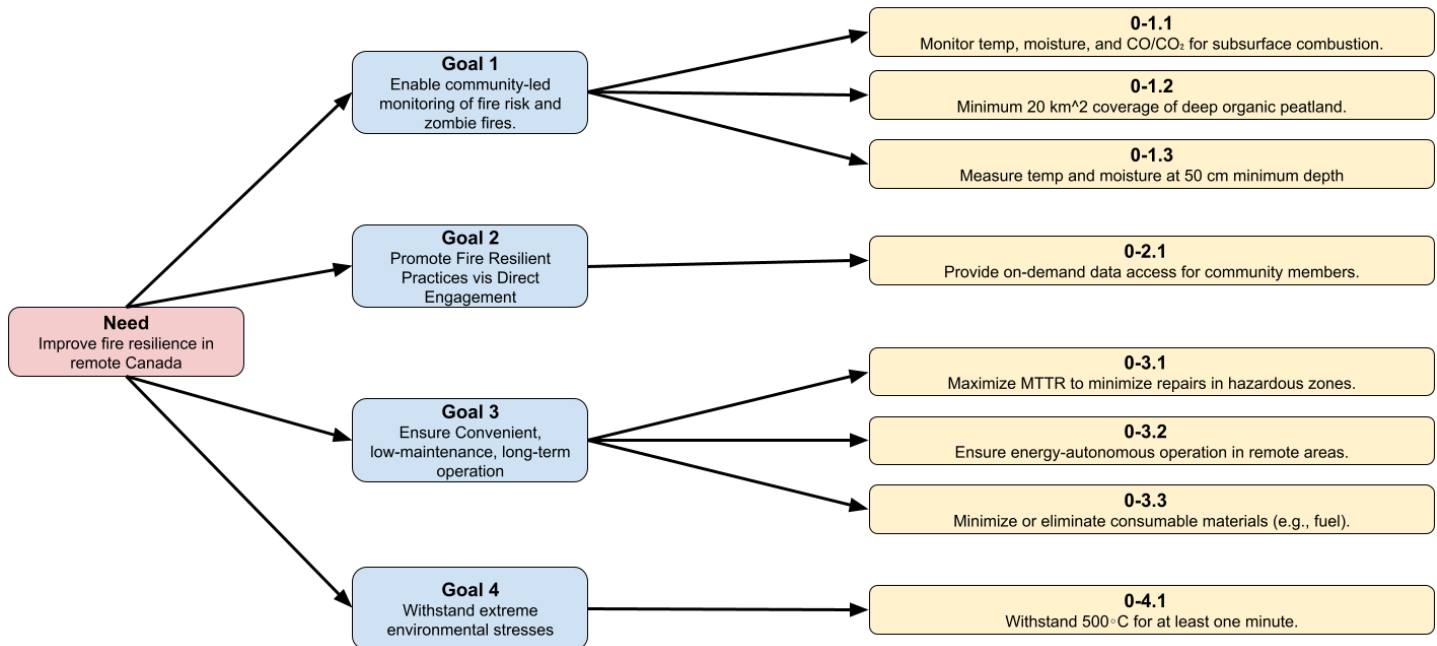


Figure 2: Updated NGOs informed by contextual and community research.

3.2 Diverging (Click: [Diverging Log](#))

The diverging process began with a clearly defined core objective: detecting overwintering “zombie fires” in peatland environments [3]. Three primary detection domains were identified: (1) subsurface, (2) aerial, and (3) satellite-based sensing.

Early ideation sessions focused on mechanical methods for implanting sensors up to 1 m below the surface while minimizing long-term maintenance requirements. Brainwriting 6-3-5 (Figure 3) was used to systematically generate concepts, while biomimicry drew inspiration from peatland fauna and natural soil interactions. The SCAMPER method was later applied to refine and hybridize promising concepts (see Diverging Log for details). Proposed subsurface solutions included a stake-based module (Fig. 3.3), a screw-driven insertion device (Fig. 3.5), and automated digging mechanisms. These concepts explored integration of temperature, moisture, gas (CO/CO₂), and GPS sensors within a structurally durable underground unit [3]. Key design constraints included structural robustness, deployability without heavy external tools, battery life exceeding one year, and protection of electronics in wet, thermally variable environments.

Subsequent sessions expanded the scope to aerial and remote sensing strategies, including drone-mounted gas and thermal sensors and satellite platforms such as NASA FIRMS [3]. Although satellite imagery was evaluated, it was not pursued as the primary solution because it does not provide direct, localized subsurface measurements of temperature, moisture, or gas concentrations [3]. Instead, satellite data was considered a complementary tool for identifying historical burn scars and informing deployment locations. Through structured ideation and iterative refinement, the team converged on hybrid concepts, such as combining rover-based mobility with a screw-driven insertion mechanism, to enhance terrain adaptability and automation [3].

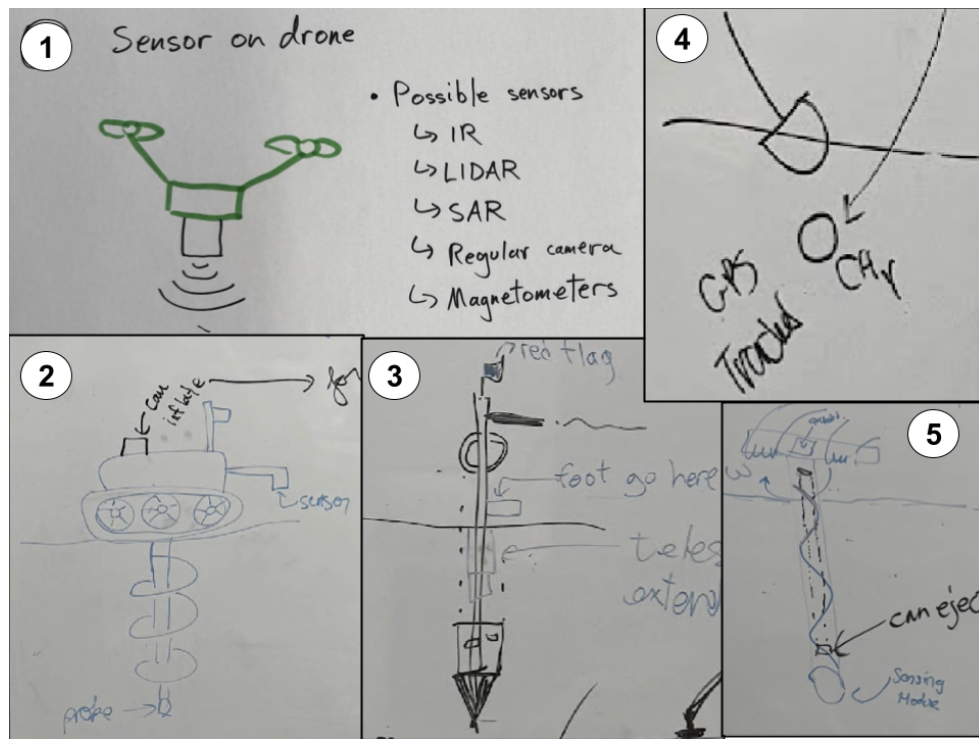


Figure 3: Brainwriting 6-3-5 session illustrating early-stage divergent concepts for subsurface zombie fire detection. (1) Drone Sensing (2) D-bot (3) Stake Sensor, (4) Ball Sensor (5) Screw Sensor.

3.3 Converging (For details: [Converging Log](#), [Converging Video](#))

Initial concepts included the ball sensor, stake, screw, D-bot, drone sensing, and satellite. Across comparisons, two critical metrics consistently dominated: (1) direct monitoring of subsurface variables (temperature, moisture, CO, CO₂) (O.1-1), and (2) maintenance feasibility and energy demand (O.3-1, O.3-2, O.3-3) [3]. Drone and D-bot concepts provided superior spatial coverage and communication, but were eliminated

Concept	Description
Ball sensor	Compact spherical module containing the required sensors and electronics, designed to be manually buried underground.
Stake sensor	Vertical probe-like structure designed to be manually inserted into peatland soil through a sharp tip and applied pressure.
D-bot	Mobile drilling platform designed to autonomously traverse peatland terrain and deploy underground sensor modules.
Screw	Rotational insertion mechanism enabling controlled placement and removal of subsurface sensors.
Drone sensing	Thermal imaging combined with CO ₂ and CO detection; drone operates at low altitude above the surface.
Satellite	Use of machine learning with thermal and radar signals to infer subsurface fire likelihood (exploratory concept).

Table 1: Primary concept directions identified during the diverging phase.

due to high energy consumption, frequent maintenance, fire vulnerability, and limited depth sensing capability [3]. Satellite sensing was similarly removed because it could not reliably monitor underground conditions and posed extreme maintenance and feasibility limitations [3].

Subsequent pairwise comparisons consistently favored the screw-based underground sensor. Compared to the stake, the screw improved ease of installation and retrieval (O.3-1) and reduced maintenance burden (O.3-2) [3]. When compared to mobile or hybrid traversal systems, the team determined that added movement complexity and energy consumption were not justified given the core requirement of continuous subsurface monitoring [3]. The screw design therefore best satisfied the critical metric of underground sensing while remaining durable, energy-efficient, and operationally practical. It is positioned as a complementary, ground-truth system that can augment satellite platforms such as NASA FIRMS rather than replace them [3].

4 Final Prototype System

Main Idea: Distributed soil probes detect subsurface smouldering indicators (heat, combustion gases, drying) and transmit data wirelessly.

These probes can be used for (1) gathering data on subsurface conditions, (2) issuing early warnings when readings indicate elevated risk, and (3) providing much-needed data to advance zombie fire research. After deciding on the Screw Sensing Module, our team held a focused discussion to refine the concept ([Link to Video](#)). The key features were finalized, and corresponding sketches and CAD designs were developed (Figure 4).

Key Features

A. Insertion and Removal: The device is easily inserted into high-alert peatlands by wildfire crews through a twisting motion (O-3.2). Figure 4.4

B. Multi-Layer Sensing: Sensors positioned at 3–4 vertical depths measure temperature, CO₂, CO, and moisture to detect subsurface smouldering gradients, with the lowest sensing layer reaching ≥ 50 cm (O-1.1).

C. Wireless Communication: Distributed modules transmit data at one-minute intervals from high-risk peatlands, enabling stakeholder access while minimizing energy use (O-1.2, O-2.1, O-3.3). The data enables education of Zombie Fires (O-2.2).

D. Lifetime: Internal battery capacity supports at least 12 months of operation (O-3.2). The battery is placed inside the cover Figure 4.3 and can be replaced.

E. Serviceability: A detachable top housing allows electronics inspection, and integrated GPS enables retrieval (O-3.1). Could also be used to replace battery and sensors.

F. Thermal Resistance: Various materials, such as metals or polytetrafluoroethylene casing, withstand temperatures approaching 500°C (O-4.1).

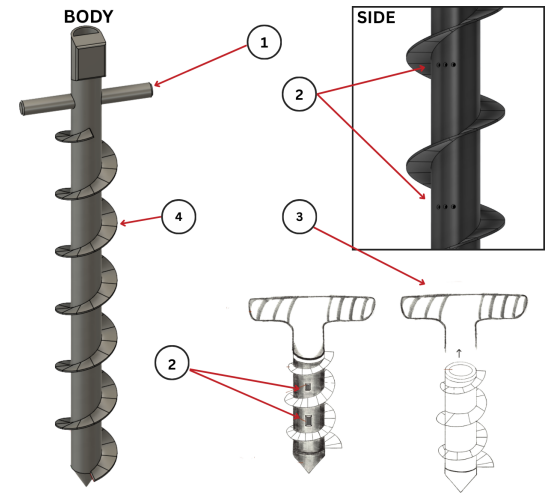


Figure 4: (1) Handles for Twisting, (2) Multi-levelled Sensors (3) Detachable Top (4) Screw Shaped Body

System Architecture

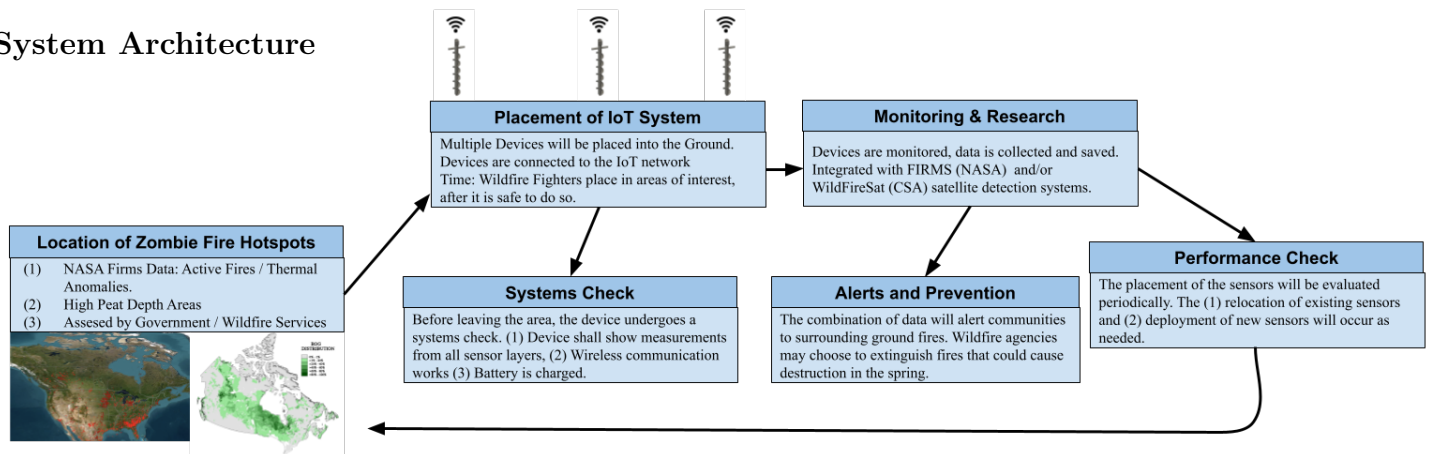


Figure 5: The figure shows the process of deploying the sensors and integrating them with existing detection systems such as FIRMS and WildFireSat. The process is cyclical and occurs repeatedly.

Future Prototypes and Direction [Click: Prototype Concept](#)

Some potential first prototypes include: (1) Gathering data from multiple sensors at different depths concurrently; (2) Screw-based digging functionality; (3) Wireless transmission of data.

References

- [1] University of Toronto Faculty of Applied Science & Engineering. *ESC204 – Praxis III: Request for Proposals (RFP), Wildland Fire Protection & Mitigation in a Canadian Context*. Version 1.0, Jan. 26, 2026.
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